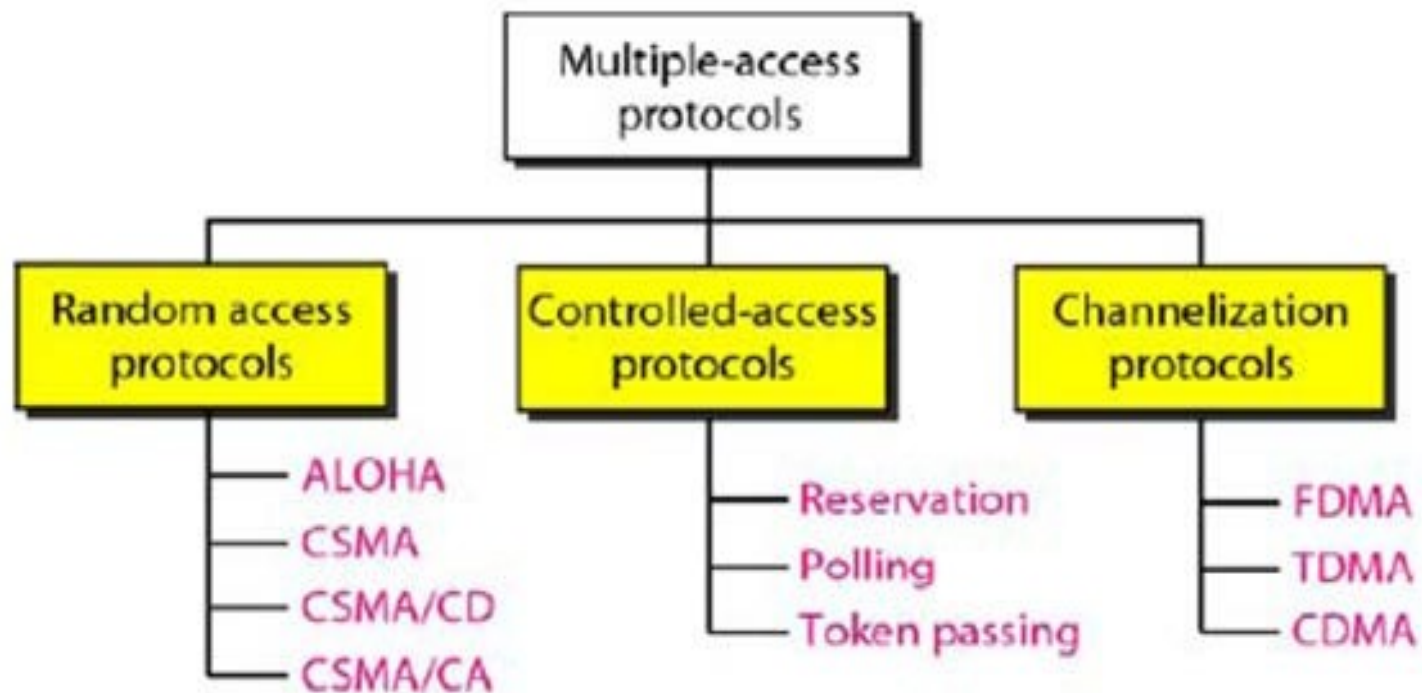

Mutiple access/Media Access protocols

- There are two types of network links: point-to-point links and broadcast links.
- A point-to-point link consists of a single sender at one end of the link and a single receiver at the other end of the link.
- Eg: link-layer protocols for point-to-point links; the point-to-point protocol (PPP)
- A broadcast link, can have multiple sending and receiving nodes all connected to the same, single, shared broadcast channel, The term broadcast is used because when any one node transmits a frame, the channel broadcasts the frame and each of the other nodes receives a copy. Eg: Ethernet and wireless LANs

- Computer networks have protocols - so-called multiple access protocols - by which nodes regulate their transmission into the shared broadcast channel.
- Multiple access protocols are required in a wide variety of network settings, including both wired and wireless local area networks, and satellite networks.
- Because all nodes are capable of transmitting frames, more than two nodes can transmit frames at the same time. When this happens, all of the nodes receive multiple frames at the same time; that is, the transmitted frames collide at all of the receivers.
- Normally, when there is a collision, none of the receiving nodes can make any sense of any of the frames that were transmitted.

- In this way, all the frames involved in the collision are lost, and the broadcast channel is wasted during the collision interval. Clearly, if many nodes want to transmit frames frequently, many transmissions will result in collisions, and much of the bandwidth of the broadcast channel will be wasted.
- In order to ensure that the broadcast channel performs useful work when multiple nodes are active, it is necessary to somehow coordinate the transmissions of the active nodes. This coordination job is the responsibility of the **multiple access protocol**.

Taxonomy of Multiple-Access Protocols



Random Access

Protocols (Contention)

- In random access or contention methods, no station is superior to another station and none is assigned the control over another.
- At each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send. (depends on the state of the medium)
- There is no scheduled time for a station to transmit. Transmission is random among the stations (*random access*)
- *Second, no rules specify which station should send next. Stations compete with one another to access the medium. That is why these methods are also called contention methods.*
- In a random access method, each station has the right to the medium without being controlled by any other station. However, if more than one station tries to send, there is an access conflict—collision—and the frames will be either destroyed or modified

Random Access

● Protocols

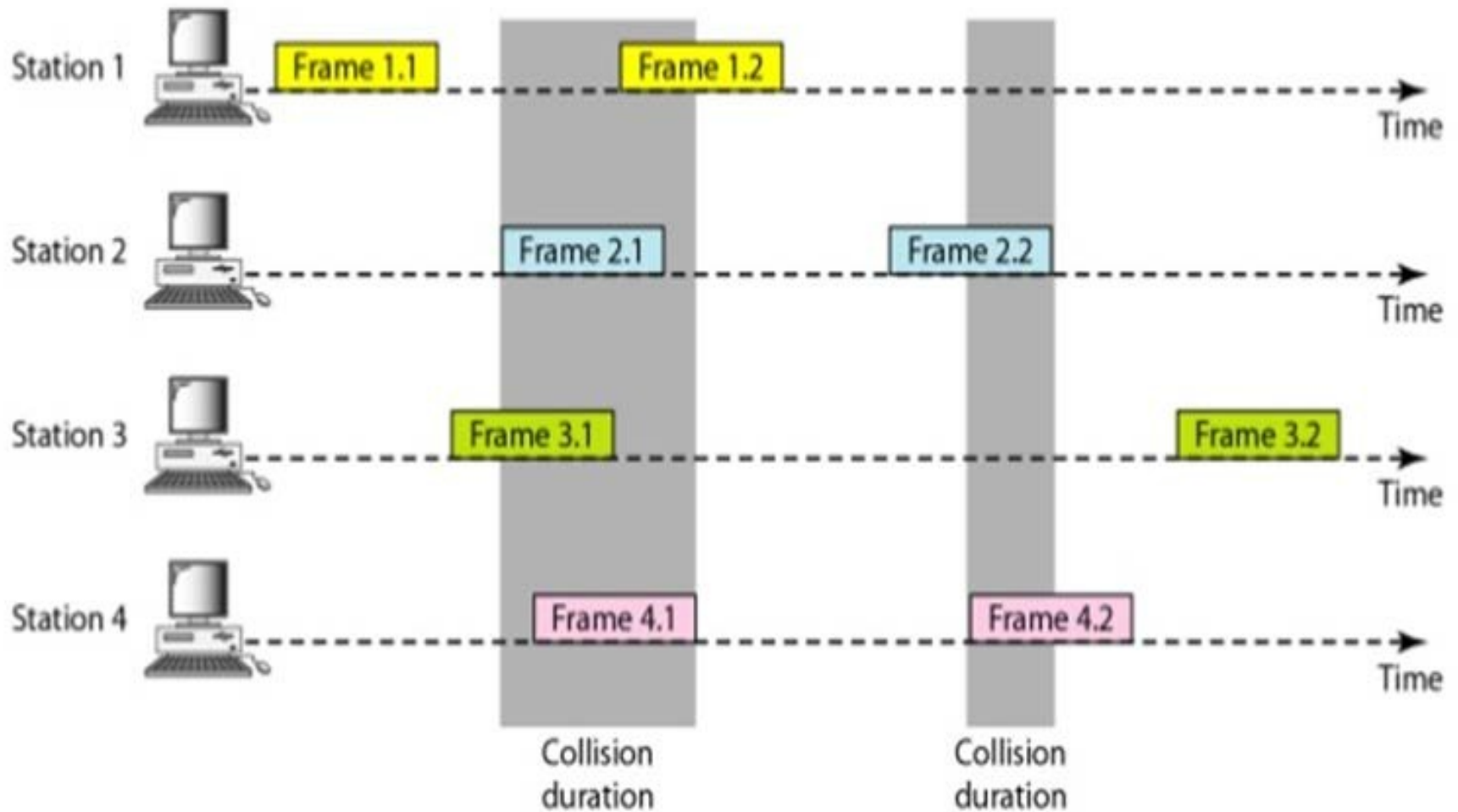
- Random access methods have evolved from a protocol called ALOHA which uses a simple procedure called Multiple Access(MA)
- This method was improved with the addition of a procedure that forces a station to sense the medium before transmitting. This is called CSMA.
- This method later evolved into two other methods CSMA/CA and CSMA/CD

ALOH

A

- Developed at the University of Hawaii for packet radio networks, applicable to any shared medium
- Pure ALOHA-original ALOHA protocol
 - Station can transmit frame at any time
 - Then listens for an amount of time equal to the maximum possible round trip propagation delay
 - If an ACK come during that time, OK, else resends the frame
 - If a station fails to receive ACK after repeated transmissions, it gives up
 - If two stations transmits at a time, **collision**
 - If the received frame is invalid, simply ignores

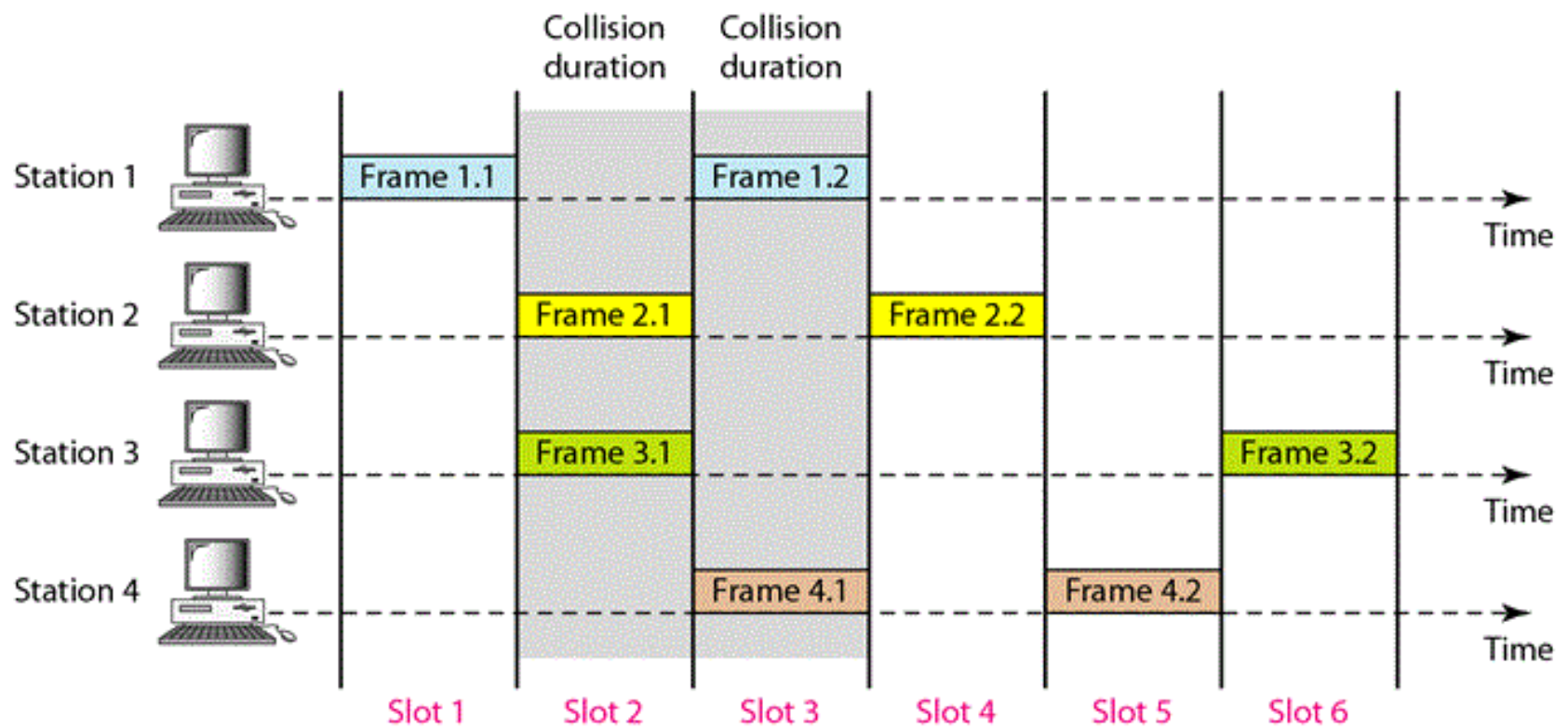
Frames in a pure ALOHA network



- We need to resend the frames that has been destroyed during transmission .
- The pure ALOHA relies on acknowledgements from the receiver.
- After sending a frame the sender waits for acknowledgement and if it does not receive ,then after timeout period it resends the frame.
- A collision involves two or more stations .If all these stations tries to resend their frames after the time out the frames will collide again.
- Pure ALOHA dictates that when the time out period passes each station waits for random amount of time before resending the frame. The randomness avoids further collisions.

Slotted ALOHA

- In Pure ALOHA there is no rule that defines when a station can send. A station may send soon after another station has started or soon after another station has finished.
- Slotted ALOHA was invented to improve the efficiency of pure ALOHA.
- In slotted ALOHA we divide the time into slots and force the station to send only at the beginning of the time slot.



- Because a station is allowed to send only at the beginning of a time slot , if a station misses this moment it should wait until the beginning of the next slot
- But there is still a possibility of collision ,if two stations tries to send at the beginning of the same time slot.

Slotted ALOHA

Assumptions

- all frames same size
- time is divided into equal size slots, time to transmit 1 frame
- nodes start to transmit frames only at beginning of slots
- nodes are synchronized
- if 2 or more nodes transmit in slot, all nodes detect collision

Operation

- when node obtains fresh frame, it transmits in next slot
- no collision, node can send new frame in next slot
- if collision, node retransmits frame in each subsequent slot

Slotted ALOHA

- Time on the channel is organized into uniform slots
- Size of slots is equal to the frame transmission time
- Central clock is used for synchronize stations
- Transmission is permitted to begin only at time slot boundary
- Frames overlaps either totally or not.
- Utilization increases to 37% than 18% of Pure ALOHA
- Poor utilization of channel capacity in ALOHA methods.

Carrier Sense Multiple Access

- A station wishing to transmit first listens the medium to determine another transmission is in progress.
- This process is called **carrier sense**.
- If the medium is in use, **station will wait** else station may **transmit**.
- If more than one station attempt to transmit at about the same time, **Collision**.
- After transmitting data, source station waits reasonable amount of time.
- Waiting time will be maximum round-trip propagation delay + time required by destination station to get medium
- If there is no Ack, collision hence retransmission

What happens if station finds the medium is busy

(1) Non persistent CSMA:

A station wishing to transmit listens to the medium and obeys the following rules

1. If the medium is idle, transmit else go to step 2
2. If the medium is busy , wait an amount of time and repeat step 1

(2) 1-Persistent CSMA:

A station wishing to transmit listens to the medium and obeys the following rules

1. If the medium is idle ,transmit else go to step 2
2. If the medium is busy , continue to listen until the channel is sensed idle, then transmit immediately

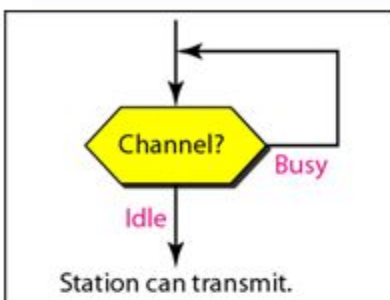
(3) P-persistent

This method is used when channel has time slots such that the time slot duration is equal to or greater than the maximum propagation delay time.

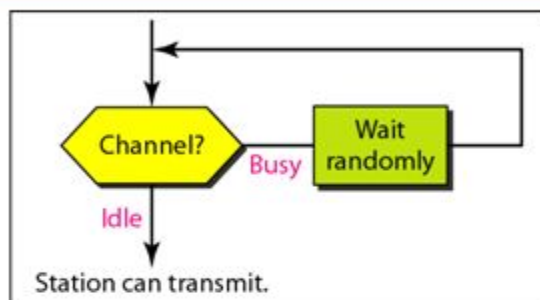
- Whenever a station becomes ready to send, it senses the channel.
- If channel is busy, station waits until next slot.
- If channel is idle, it transmits with a probability p .
- With the probability $q=1-p$, the station then waits for the beginning of the next time slot.
- If the next slot is also idle, it either transmits or waits again with probabilities p and q .

This process is repeated till either frame has been transmitted or another station has begun transmitting. In case of the transmission by another station, the station acts as though a collision has occurred and it waits a random amount of time and starts again.

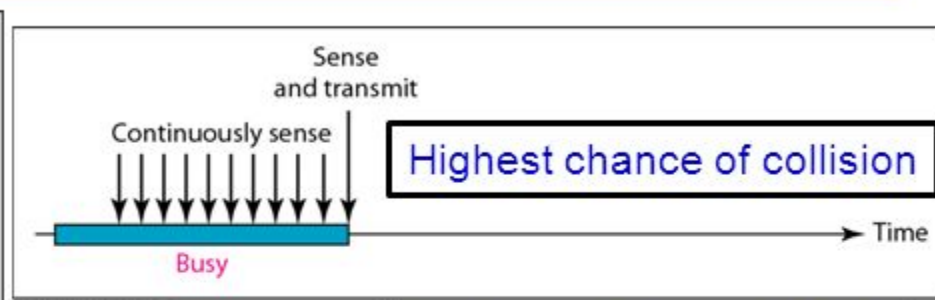
Flow diagram for three persistence methods



a. 1-persistent

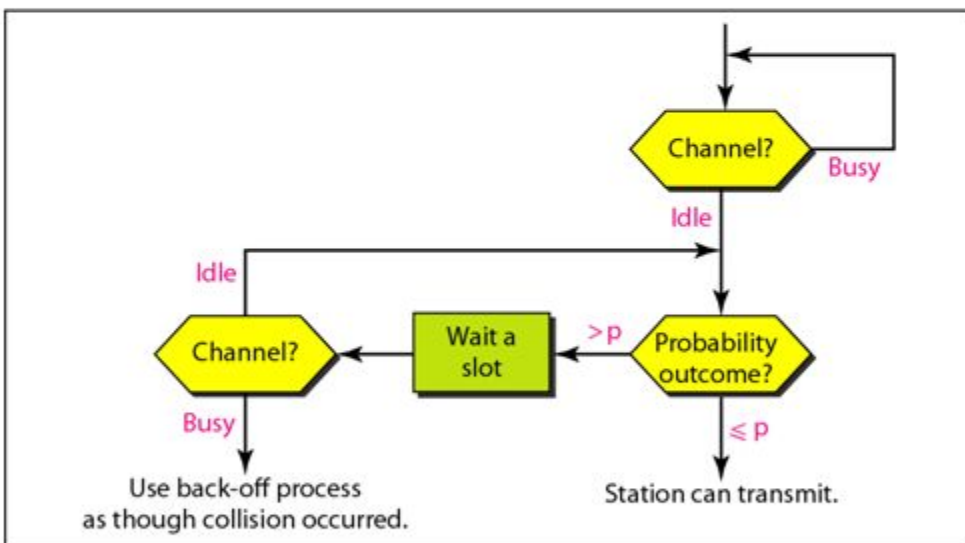


b. Nonpersistent

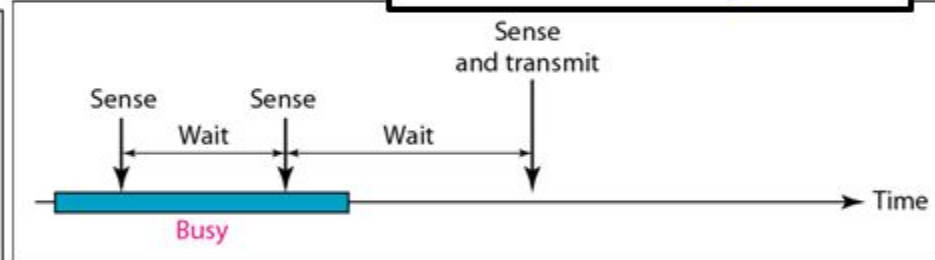


a. 1-persistent

Highest chance of collision

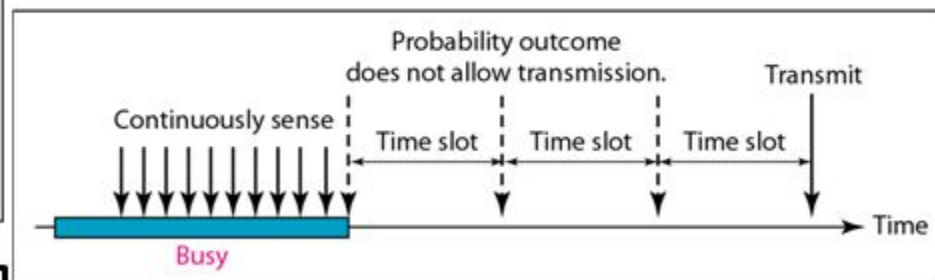


c. p-persistent



b. Nonpersistent

Reduce efficiency of N/W

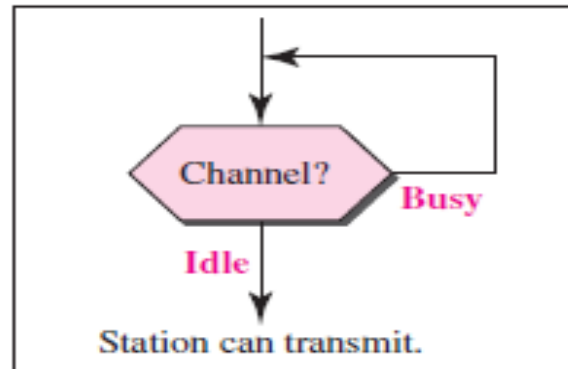


c. p-persistent

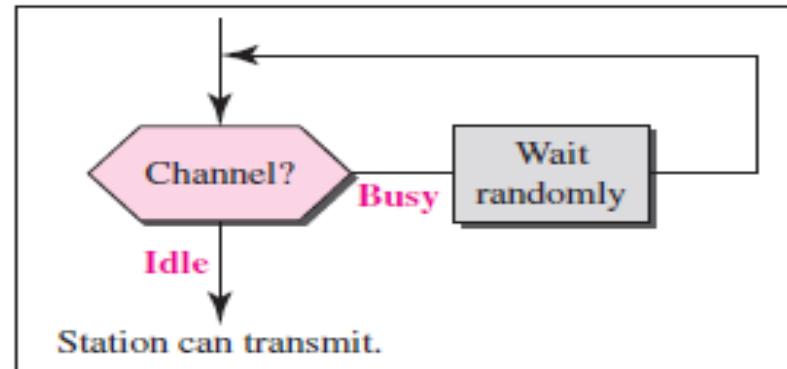
Behavior of three persistence methods

Channel has time slot : $\geq$ Max. propagation time
 Combine advantages of two methods
 - Reduce chance of collision
 - improve efficiency

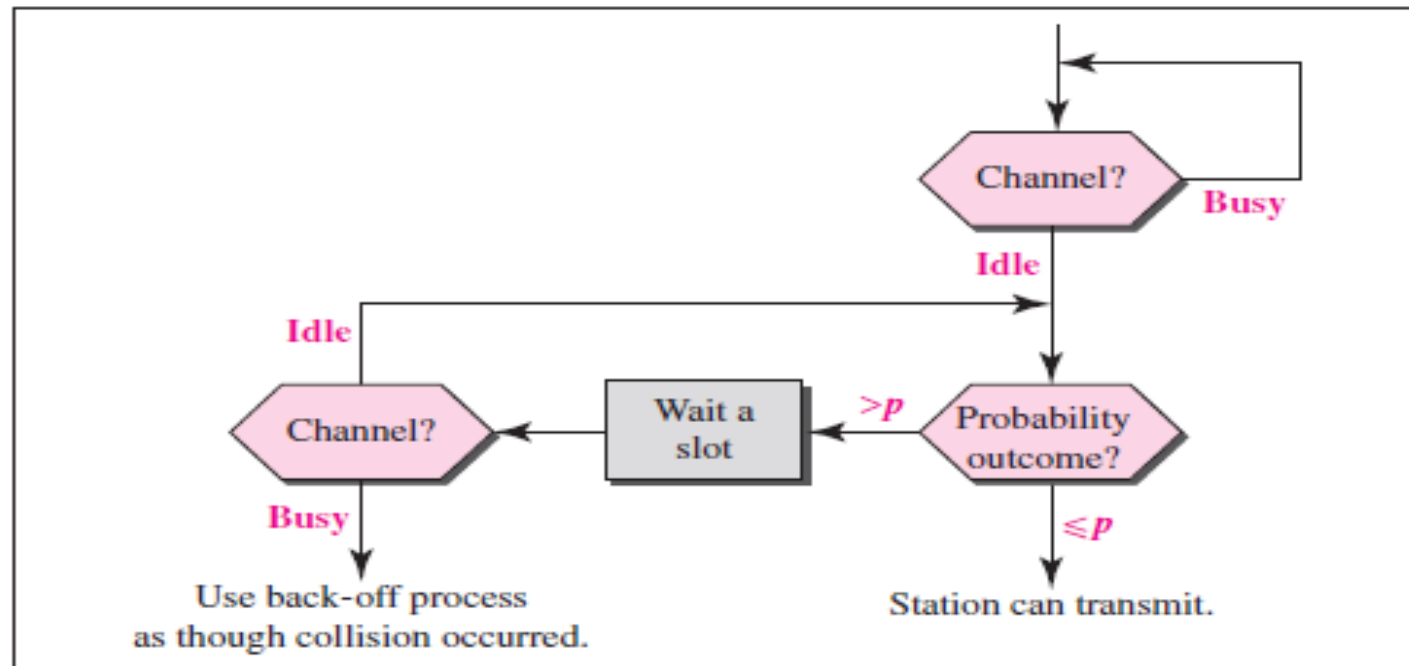
Flow diagram for three persistence methods



a. 1-persistent



b. Nonpersistent

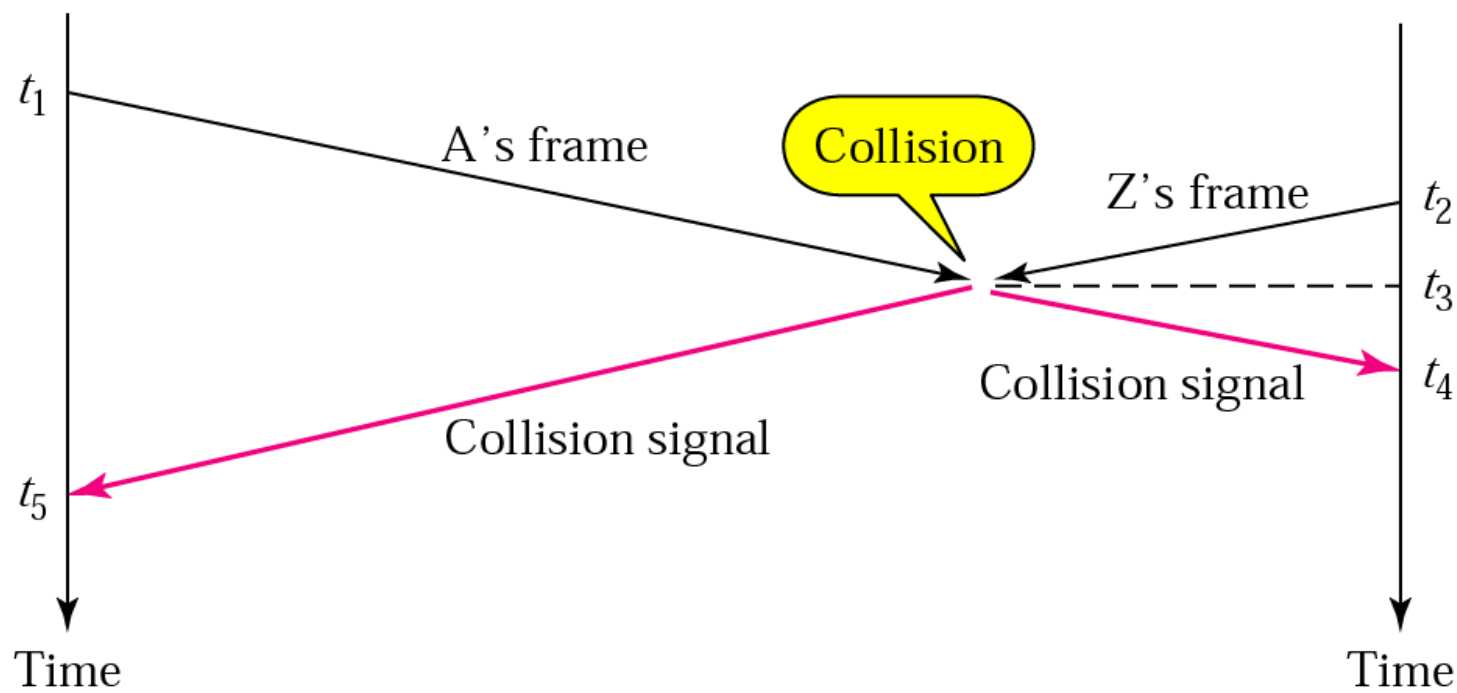
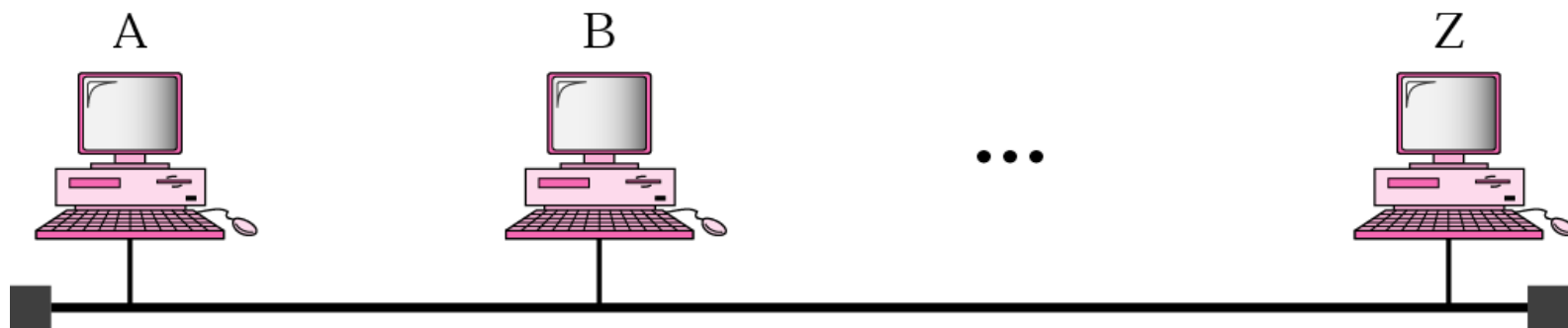


c. p -persistent

CSMA/

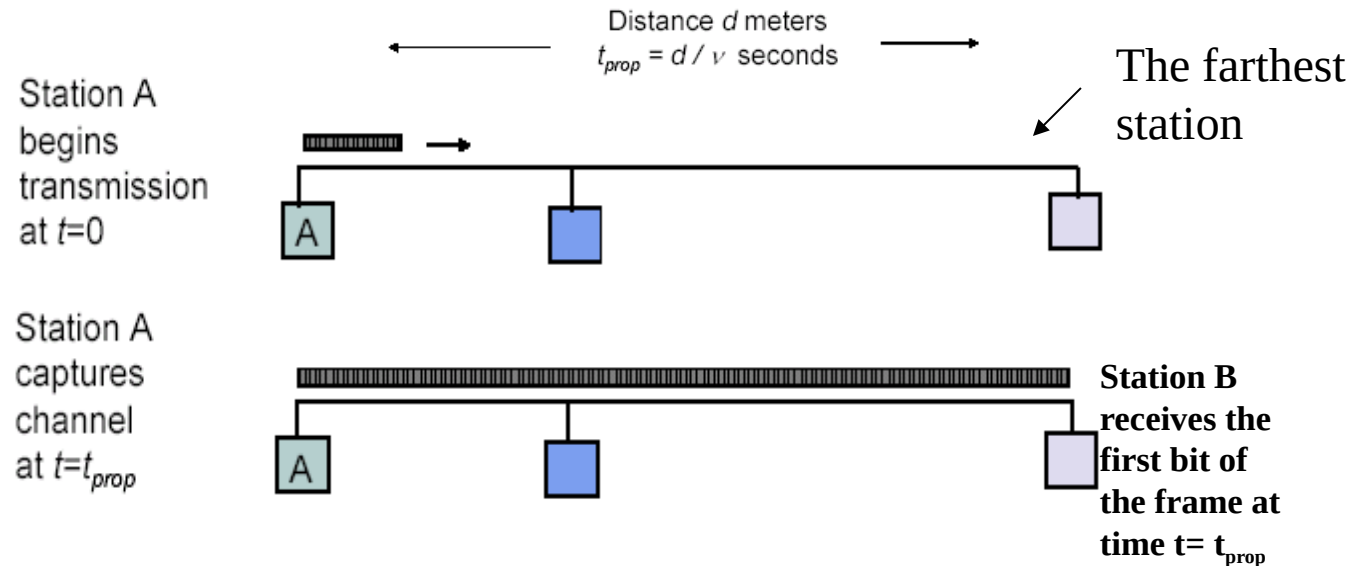
CD

- When two frames collide, the medium remains unstable for the duration of transmission of both damaged frames
- Hence an alternate method, where
 - A station continues to listen to the medium while transmitting
- Rules of CSMA/CD
 - If the medium is idle, transmit
 - If the medium is busy, continue to listen until the channel is idle; then transmit immediately
 - If a collision is detected during transmission, transmit a brief jamming signal to assure that all stations know that there is a collision and then cease transmission
 - After transmitting jamming signal, wait a random amount of time, referred to as **backoff** then attempt to transmit again

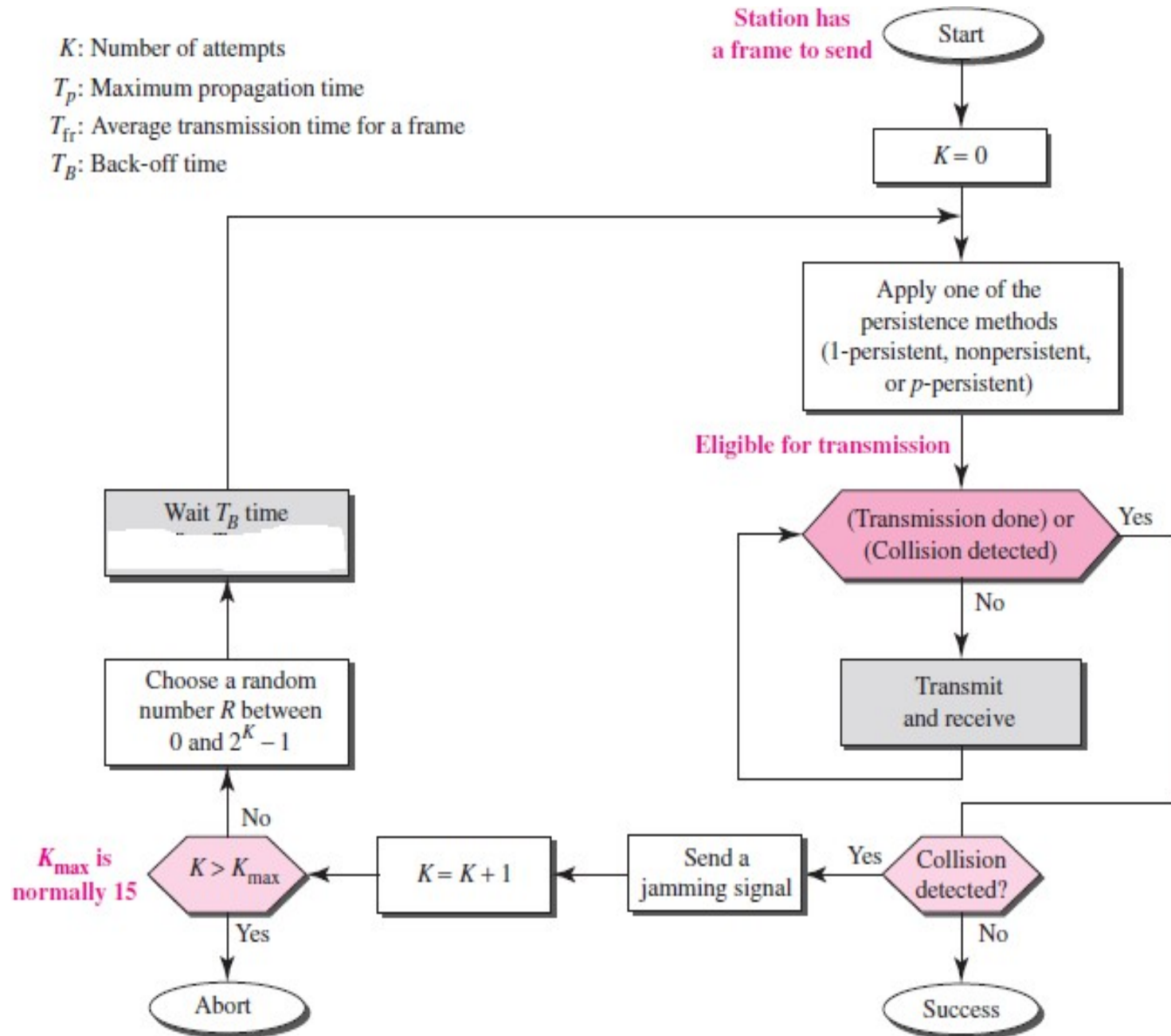


Maximum Propagation Delay

Maximum propagation delay(t_{prop}): time it takes for a bit of a frame to travel between the **two most widely** separated stations.



K : Number of attempts
 T_p : Maximum propagation time
 T_{fr} : Average transmission time for a frame
 T_B : Back-off time



Exponential Backoff

- Backoff is the process by which a transmitting station determines how long to wait following a collision before attempting to retransmit the frame.
- If all stations waited the same length of time before retransmission, then another collision would inevitably occur.
- This is avoided by having each station generate a random number which determines the length of time it must wait before testing the carrier. This time period is known as the station's "**backoff delay**".

CSMA/

CA

- CSMA/CD is detecting collisions while transmitting data
- Collisions can be detected with the help of energy of the signal
- In wired network, energy of the signal is almost double when collision occurs, hence easy detection
- But in wireless communication, energy level may not increase to a considerable level when collision occurs, hence less chance to detect collisions
- Collision avoidance is better method than collision detection in these type of networks

- Collisions are avoided through the use of CSMA/CA's three strategies: the interframe space, the contention window, and acknowledgments.

- ***Interframe Space (IFS)***

- First, collisions are avoided by deferring transmission even if the channel is found idle. When an idle channel is found, the station does not send immediately. It waits for a period of time called the **interframe space or IFS**.

- **Even though the channel may** appear idle when it is sensed, a distant station may have already started transmitting.

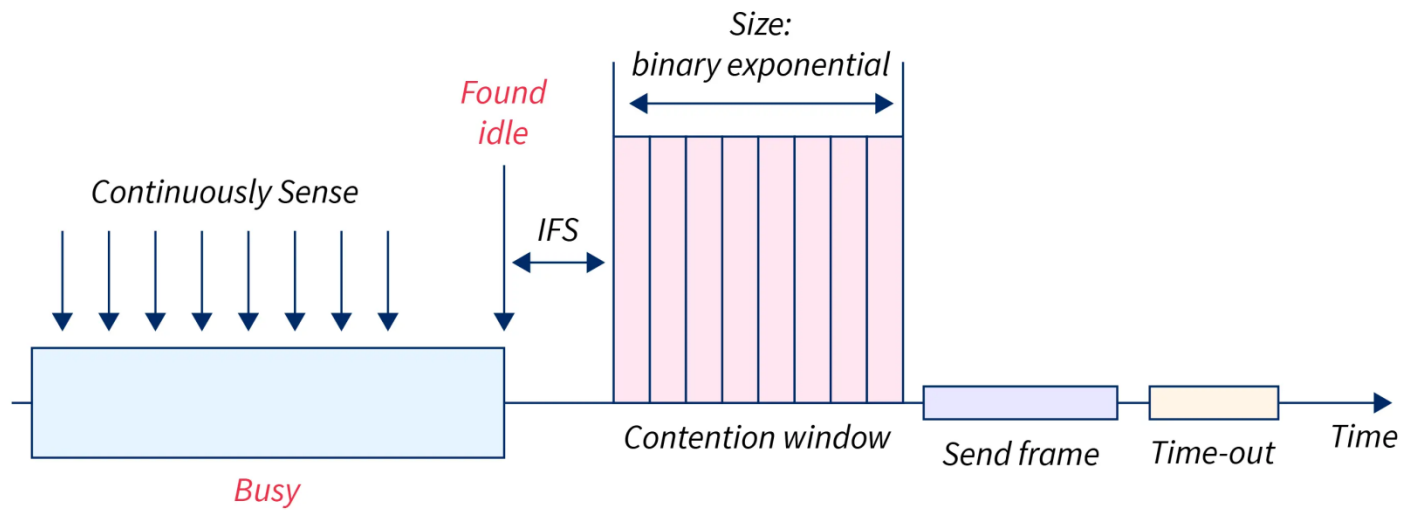
● ***Contention Window***

- The contention window is an amount of time divided into slots. A station that is ready to send chooses a random number of slots as its wait time.
- It is set to one slot the first time and then doubles each time the station cannot detect an idle channel after the IFS time.
- However, if the station finds the channel busy, it does not restart the process; it just stops the timer and restarts it when the channel is sensed as idle. This gives priority to the station with the longest waiting time.

Acknowledgment

- The positive acknowledgment guarantee that the receiver has received the frame.

Timing in CSMA/CA

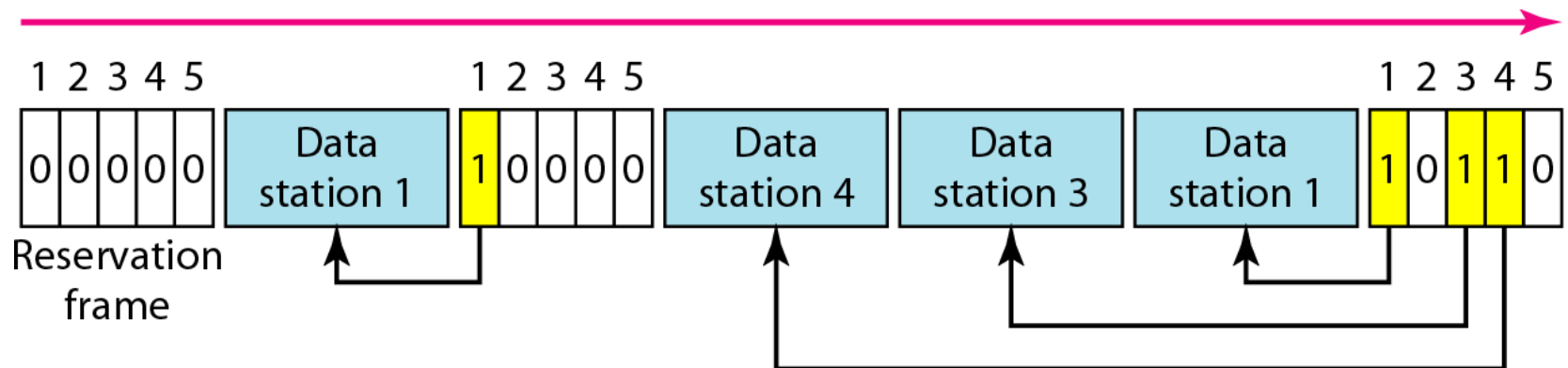


Controlled Access

- Stations consult one another to find which station has the right to send
- A station cannot send unless it has been authorized by other stations
- Three controlled access methods
 - **Reservation**
 - **Polling**
 - **Token Passing**

1-Reservation access method

- Stations **take turns transmitting a single frame** at a **full rate (R) bps**
- Transmissions are organized into variable length cycles
- Each cycle begins with a reservation interval that consists of (N) minislots. One minislot for each of the N stations
- When a station needs to send a data frame, it makes a **reservation** in its own minislot.
- By listening to the reservation interval, every station knows which stations will transfer frames, and in which order.
- The stations that made reservations can send their data frames after the reservation frame.



Polling

- It works with topologies in which one device is primary station and all other stations are secondary stations
- All data exchanges must be made through primary station, even when ultimate destinations are secondary
- Primary device controls the link, secondary device follows its instruction
- Primary station determines which station can use the channel

Polling access method

- Devices are designated as a **Primary station** and **secondary station**.
- All data exchange must go through the primary station
- **Primary** station controls the link and **initiates** the session; the secondary station follow its instructions.
- Two functions: **polls** and **select**
 - **Poll** : If the primary wants to **receive data**, it asks the secondary if they have anything to send.
 - **Select**: if the primary wants to **send data**, it tells the secondary to get ready to receive data

Polling

g When primary want send or receive data, it uses Select function or Poll function

- Select

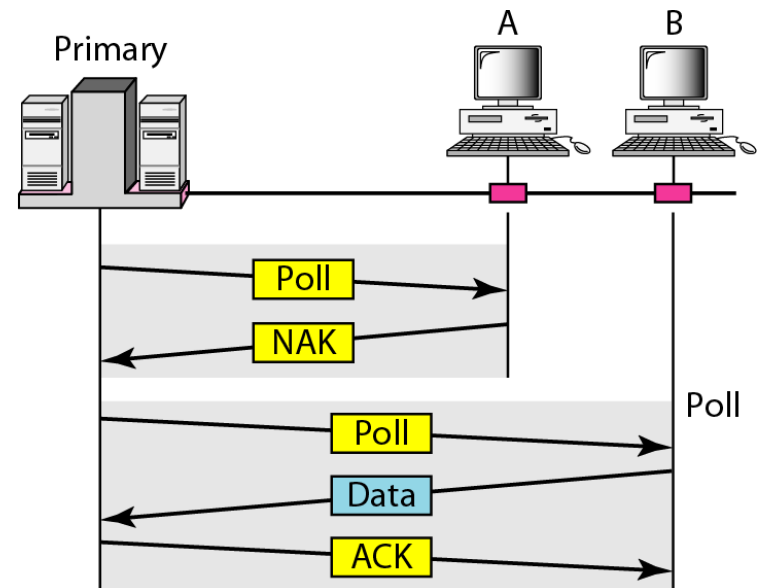
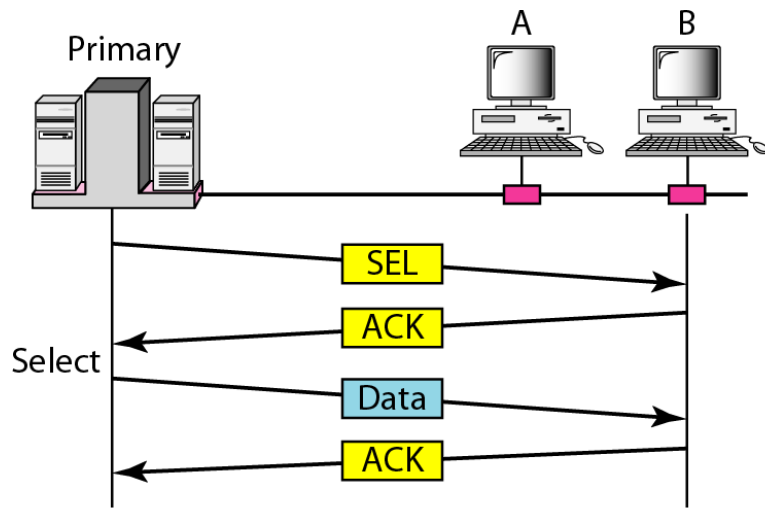
- This function is used when primary device want to send data
- If the primary is neither sending nor receiving data, medium is available. Since all data transmissions happens through primary
- Primary alerts desired secondary by sending select(SEL) frame
- It waits for acknowledgment before sending data

Polling

g Poll

- It is the process in which the primary stations asks secondary station, whether they want to send data
- Primary Polls all secondary stations in turn
- If the secondary want to send data, it responds with data frame
- Otherwise responds with NAK frame
- When data is received from secondary, it responds with ACK frame

SELECT and POLL functions



Token Passing

- In the token-passing, the stations are organized in a **logical ring**.
- For each station there is a predecessor and a successor
- predecessor is the station which is logically before the station in the ring
- The successor is the station which is logically after the station in the ring
- The current station is the one that accessing the channel now.

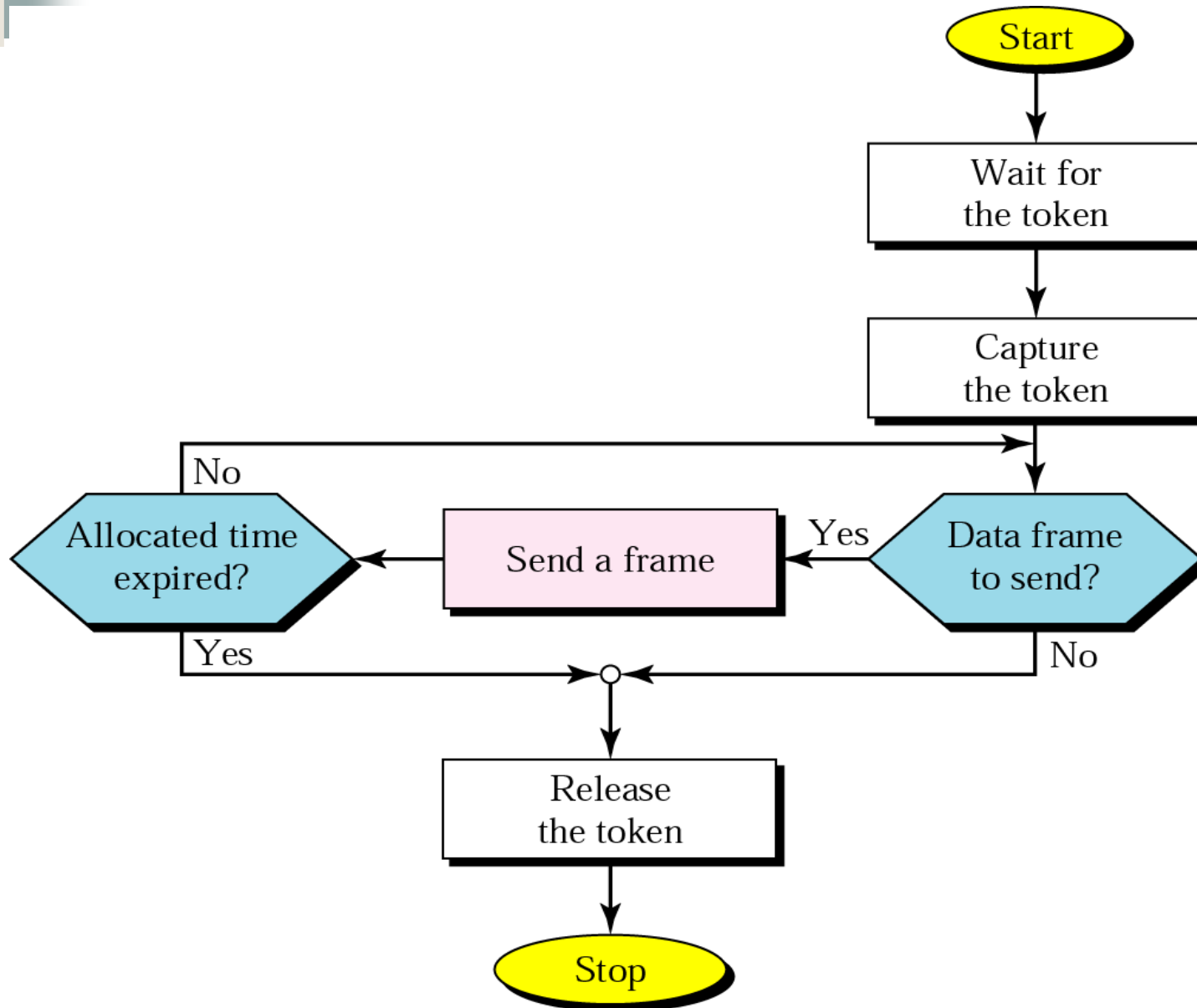
Token Passing

- Station is authorized to send data when it receives a special frame called a **token**
- Stations are arranged around a ring.
- When no data are being sent, a token circulates the ring.
- If a station needs to send data, it waits for the token.
- The station captures the token and sends one or more frames (as long as it has frames to send or the allocated time has not expired), and finally it releases the token to be used by next station (successor).
- The maximum time any station can hold the token is limited.
- Since there is only one token, only one station transmits at a time, and **collisions never occur.**

Token Passing

- **Token management is needed for this access method:**
 - Stations must be limited in the time they can have possession of the token
 - the token must be monitored to ensure it has not been lost or destroyed(if the station that is holding the token fails, the token will disappear from the network)
 - Assign priorities to the stations , to make low-priority stations release the token to high priority stations.
- Stations do not have to be physically connected in a ring ; the ring can be logical one

Token-passing procedure



❖ CHANNELIZATION PROTOCOLS

- Channelization (or channel partition) is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations.

1. Frequency-Division Multiple Access (FDMA)

- The available bandwidth is divided into frequency bands. Each station is allocated a band to send its data.
- Each band is reserved for a specific station, and it belongs to the station all the time.
- Each station also uses a band-pass filter to confine the transmitter frequencies.

CHANNELIZATION PROTOCOLS

Channelization is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, among different stations. There are 3 channelization protocols: FDMA, TDMA, and CDMA

FDMA

- In frequency-division multiple access (FDMA), the available bandwidth is divided into frequency bands.
- Each station is allocated a band to send its data.
- Each band is reserved for a specific station, and it belongs to the station all the time.
- Each station also uses a band pass filter to confine the transmitter frequencies. To prevent station interferences, the allocated bands are separated from one another by small guard bands.

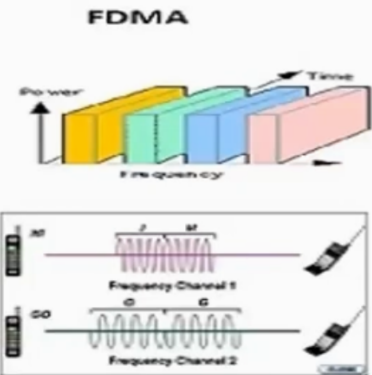
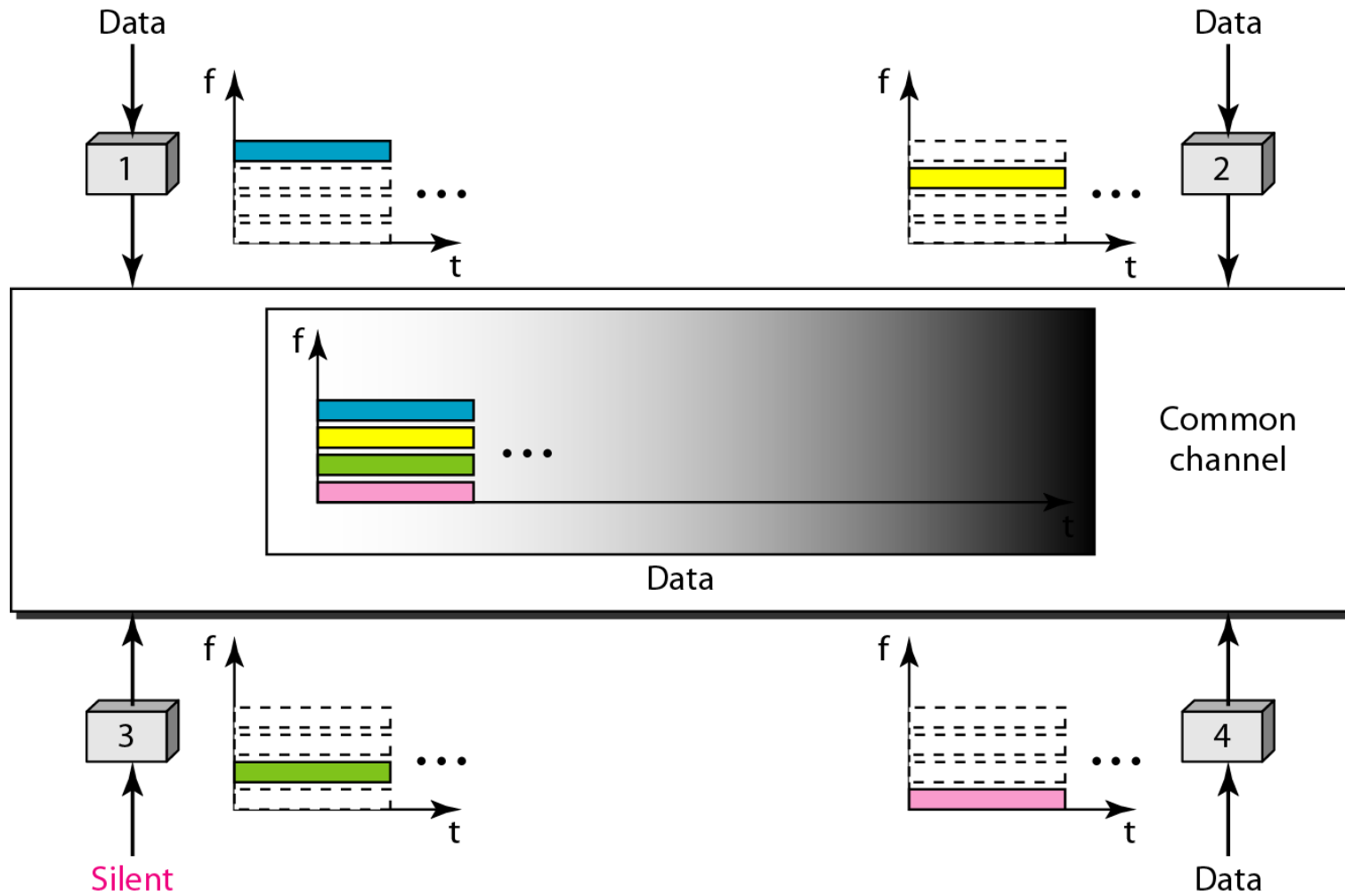


Figure 12.21 *Frequency-division multiple access (FDMA)*



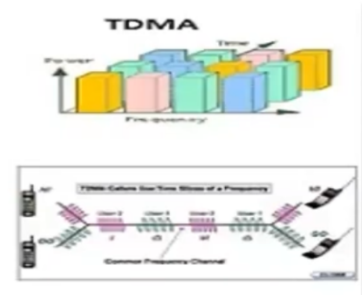
- FDMA specifies a predetermined frequency band for the entire period of communication.
- This means that stream data (a continuous flow of data that may not be packetized) can easily be used with FDMA.
- This feature can be used in cellular telephone systems

2. Time-Division Multiple Access (TDMA)

- The stations share the bandwidth of the channel in time.
- Each station is allocated a time slot during which it can send data.
- Each station transmits its data in its assigned time slot.

- **TDMA**

- TDMA, is a physical layer technique that combines the data from slower channels and transmits them by using a faster channel.
- The process uses a physical multiplexer that interleaves data units from each channel
- TDMA, is an access method in the data-link layer. The data-link layer in each station tells its physical layer to use the allocated time slot. There is no physical multiplexer at the physical layer.
- Each station is allocated a time slot during which it can send data.
- Each station transmits its data in its assigned time slot.
- The main problem with TDMA lies in achieving synchronization between the different stations. Each station needs to know the beginning of its slot and the location of its slot.



- The main problem with TDMA lies in achieving synchronization between the different stations.
- Each station needs to know the beginning of its slot and the location of its slot. This may be difficult because of propagation delays introduced in the system if the stations are spread over a large area.
- To compensate for the delays, we can insert guard times. Synchronization is normally accomplished by having some synchronization bits (normally referred to as preamble bits) at the beginning of each slot.

12-3 CHANNELIZATION - TDMA

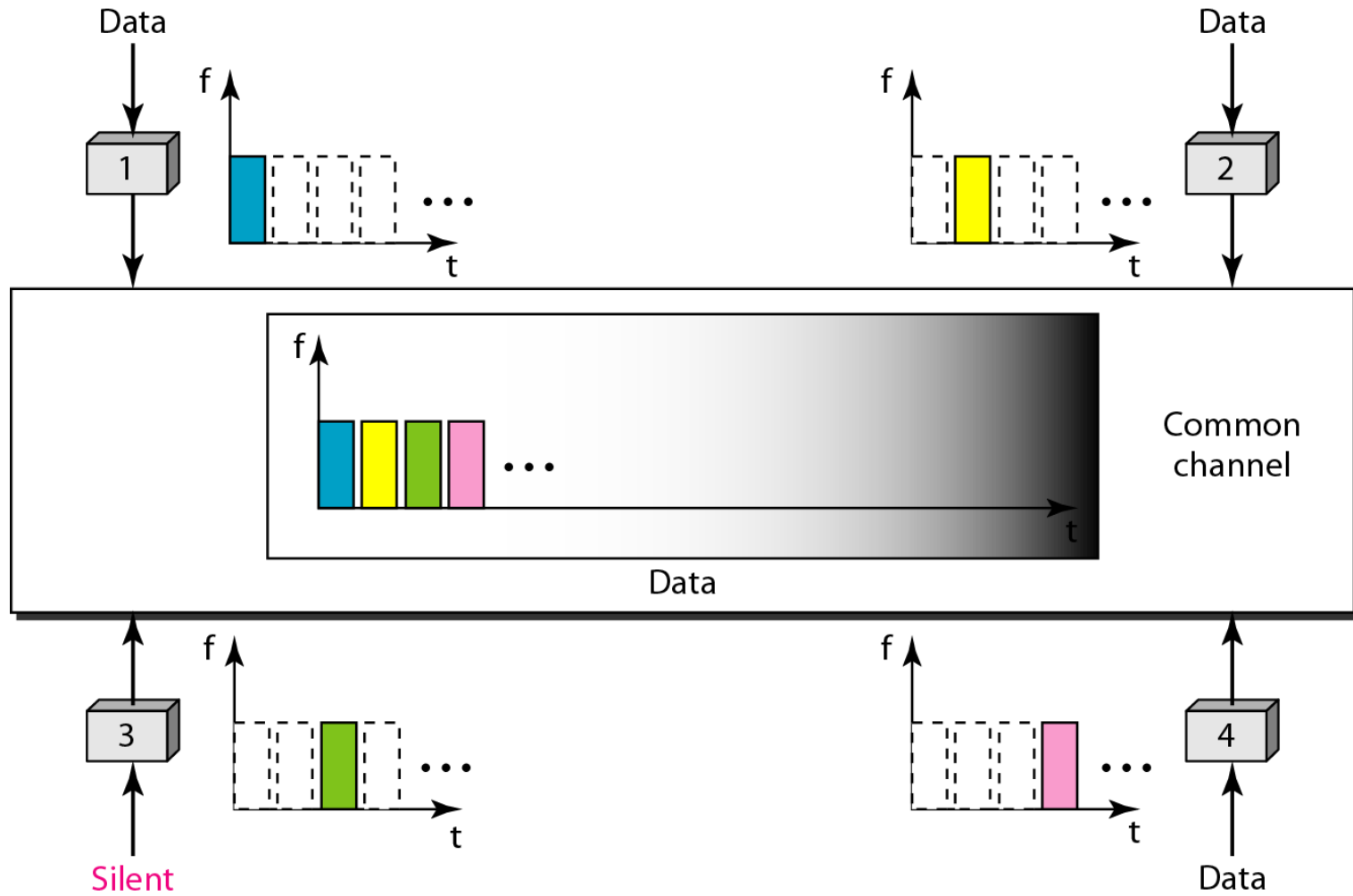
TDMA: Time Division Multiple Access

The entire bandwidth capacity is a **single channel** with its capacity shared **in time** between **M** stations

A node must **always wait for its turn** until its slot time arrives even when it is the **only node** with frames to send

A node is limited to an average rate equal **R/M** (where **M** is number of nodes) even when it is the only node with frame to be sent

Figure 12.22 *Time-division multiple access (TDMA)*



12-3 CHANNELIZATION - CDMA

- **CDMA: Code Division Multiple Access**
 - In CDMA, one channel carries all transmissions **simultaneously**
 - Each station codes its data signal by a specific codes before transmission
 - The stations receivers use these codes to recover the data for the desired station

- Let us assume we have four stations, 1, 2, 3, and 4, connected to the same channel. The data from station 1 are d_1 , those from station 2 are d_2 , and so on.
- The code assigned to the first station is c_1 , to the second is c_2 , and so on.
- The assigned codes have two properties.

1. If we multiply each code by another, we get 0.

2. If we multiply each code by itself, we get 4 (the number of stations).

- Station 1 multiplies (a special kind of multiplication) its data by its code to get $d_1 \cdot c_1$.
- Station 2 multiplies its data by its code to get $d_2 \cdot c_2$, and so on.

3. Code-Division Multiple Access (CDMA)

- CDMA differs from FDMA because only one channel occupies the entire bandwidth of the link.
- It differs from TDMA because all stations can send data simultaneously; there is no time-sharing
- CDMA simply means communication with different codes.
- For example, in a large room with many people, two people can talk in English if nobody else understands English. Another two people can talk in Chinese if they are the only ones who understand Chinese, and so on.
- In other words, the common channel, the space of the room in this case, can easily allow communication between several couples, but in different languages (codes).

CDMA

- CDMA(Code Division Multiple Access) differs from FDMA in that only one channel occupies the entire bandwidth of the link.
- It differs from TDMA in that all stations can send data simultaneously; there is no timesharing.

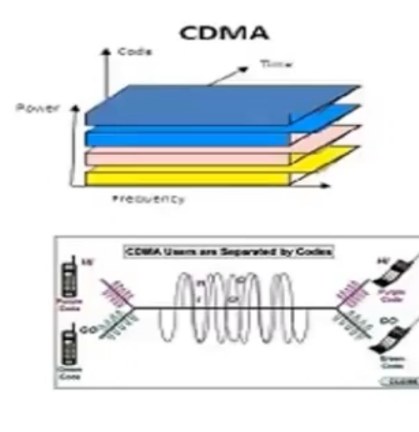
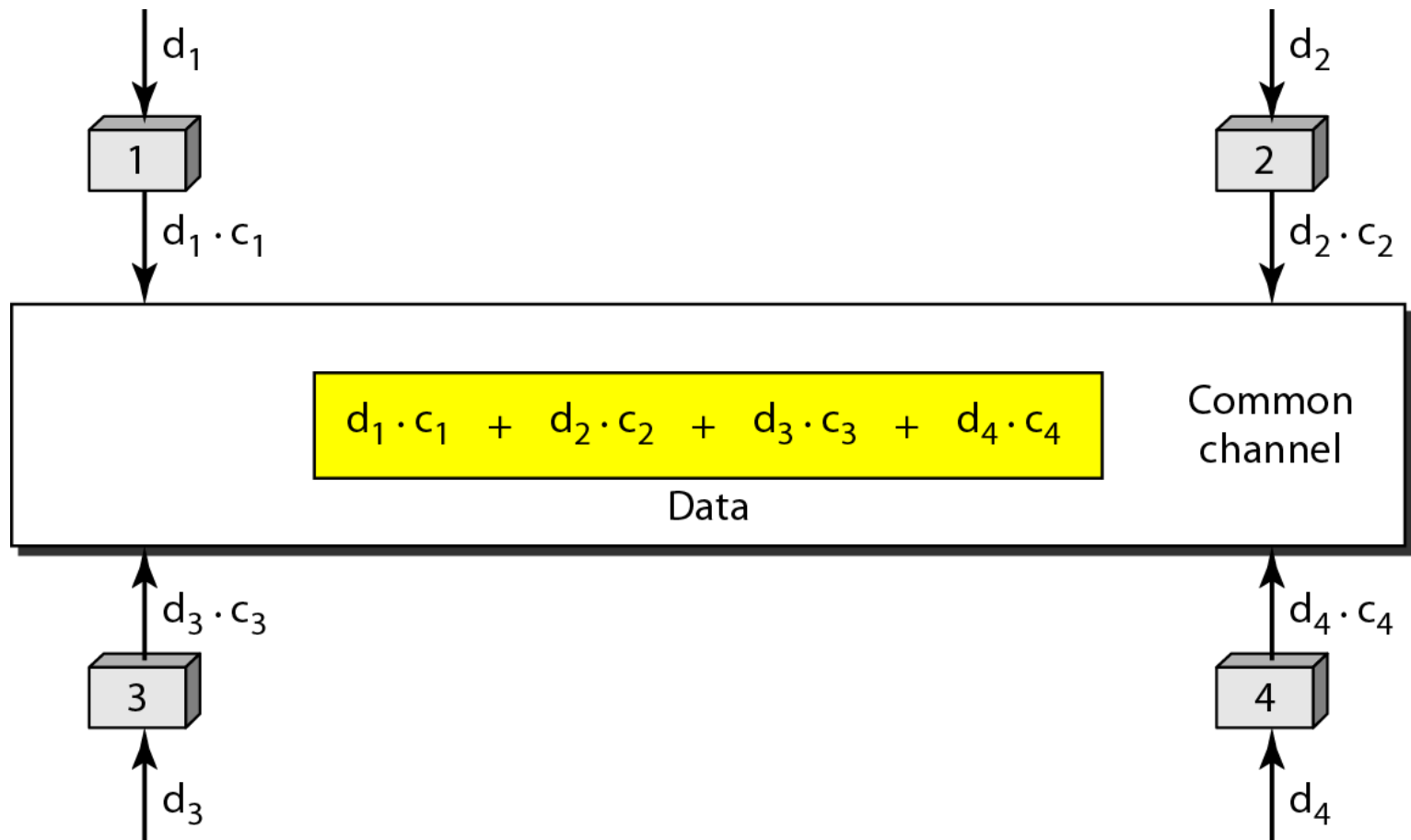


Figure 12.23 Simple idea of communication
with code



- The data that go on the channel are the sum of all these terms.
- Any station that wants to receive data from one of the other three stations multiplies the data on the channel by the code of the sender.
- For example, suppose stations 1 and 2 are talking to each other. Station 2 wants to hear what station 1 is saying. It multiplies the data on the channel by c_1 , the code of station 1.
- Because $(c_1 \cdot c_1)$ is 4, but $(c_2 \cdot c_1)$, $(c_3 \cdot c_1)$, and $(c_4 \cdot c_1)$ are all 0s, station 2 divides the result by 4 to get the data from station 1.

$$\text{data} = [(d_1 \cdot c_1 + d_2 \cdot c_2 + d_3 \cdot c_3 + d_4 \cdot c_4) \cdot c_1] / 4 = [d_1 \cdot c_1 \cdot c_1 + d_2 \cdot c_2 \cdot c_1 + d_3 \cdot c_3 \cdot c_1 + d_4 \cdot c_4 \cdot c_1] / 4 = (4 \times d_1) / 4 = d_1$$